

7) Prove the domination laws in Table 1 by showing that

a) $A \cup U = U$ b) $A \cap \emptyset = \emptyset$

Solution: a) Let, $x \in A \cup U$. By Defn. 1, $x \in A \vee x \in U$ is True. By Defn. of OR, at least one of $x \in A$ or $x \in U$ is True. If $x \in U$, we are done. Suppose, $x \in A$. We know that $A \subseteq U$. By Defn. 3 of Sec-2.1, $x \in U$ is True. $\therefore x \in U$ always holds. By Defn. 3 of Section 2.1, $A \cup U \subseteq U$, since we had assumed that x is an arbitrary element of $A \cup U$.

Let $x \in U$. By Defn. of OR, $x \in A \vee x \in U$ is True. By Defn. of union, $x \in A \cup U$. x is an arbitrary element of U . By Defn. 3 of Section 2.1, $U \subseteq A \cup U$.

$\therefore A \cup U = U \quad \square$

b) Let x be an arbitrary element of $A \cap \emptyset$, i.e. $x \in A \cap \emptyset$. By Defn. 2, $x \in A$ and $x \in \emptyset$. By Defn. of Empty Set, for any arbitrary element p , $p \in \emptyset$. $\therefore x \in \emptyset$. This is a contradiction. $\therefore$ No such x exists where $x \in A \cap \emptyset$, i.e. $x \notin A \cap \emptyset$. By defn. of conditional, $x \notin \emptyset \rightarrow x \notin A \cap \emptyset$ is True. By the contrapositive and Defn. 3 of Sec-2.1, $A \cap \emptyset \subseteq \emptyset$.

Let x be an arbitrary element of $\emptyset$, i.e. $x \in \emptyset$. By Defn. of empty set, $x \notin \emptyset$. By Defn. of not belongs to, $\neg(x \in \emptyset)$ is True. By Defn. of OR, $\neg(x \in \emptyset) \vee \neg(x \in A)$ is True. By De Morgan's Law of proposition, $\neg(x \in \emptyset \wedge x \in A)$ is True. By Commutative Law of proposition, $\neg(x \in A \wedge x \in \emptyset)$ is True. By Defn. 2, $\neg(x \in A \cap \emptyset)$ is True. By Defn. of not belongs to $x \notin A \cap \emptyset$. By defn. of conditional, $x \notin \emptyset \rightarrow x \notin A \cap \emptyset$ is True. By the contrapositive and Defn. 3 of Sec-2.1, $A \cap \emptyset \subseteq \emptyset$.

$\therefore A \cap \emptyset = \emptyset \quad \square$